GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 8

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Recap of Previous Lecture

Topic Predicate logic

Topic

Topic

Topic

Topic

Turn on Slide map

Topics to be Covered

- Topic Predicatelogic
- Topic
- Topic
- Topic
- Topic

“Don’t let yesterday take up too much of today.”

Will Rogers

Not all players are good at maths → Some are good

→ Some are not good

• None player good at maths → All are bekar.

All player are not good at math

$A \rightarrow B \Rightarrow (\neg A \vee B)$

Negation of a statement

What is the logical translation of the following statements?

"None of my friends are perfect." $\Rightarrow$ $\forall x\,(F(x) \longrightarrow \neg P(x))$

(A) $\exists x\,(F(x) \wedge \neg P(x))$

(B) $\exists x(\neg F(x) \wedge P(x))$

(C) $\exists x\,(\neg F(x) \wedge \neg P(x))$

(D) $\neg \exists x(F(x) \wedge P(x))$

$\sim\sim \forall x\,(\neg F(x) \vee \neg P(x))$

$\sim \exists x\,(F(x) \wedge P(x))$

$$\boxed{\forall x\,(\text{Predicate})} \Rightarrow$$ if Predicate is false for any x then this is false

$$\Leftarrow \boxed{\exists x\,(\text{Predicate})}$$ If the predicate is true for a single x then this is true.

Negation of a statement

Let p(x) and q(x) denote the following open statements.

p(x): $x^2 > 0$

q(x): x is odd

for the universe of all integers, determine the truth or falsity of each of the statement.

S_1: $\forall x\ [p(x) \rightarrow q(x)]$

S_2: $\exists x\ [p(x) \rightarrow q(x)]$

Which of the following is true?

(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) Neither S_1 nor S_2

meaning of Implies

S_1: $\forall x\ (P(x) \rightarrow q(x))$ ⇒ false

- for all x where P(x) is true it implies that q(x) is also true

@x=2 P(2) is true but q(2) is false

Negation of a statement

Let p(x) and q(x) denote the following open statements.

p(x): $x^2 > 0$

q(x): x is odd

for the universe of all integers, determine the truth or falsity of each of the statement.

S_1: $\forall x\,[p(x) \rightarrow q(x)]$ ← False

S_2: $\exists x\,[p(x) \rightarrow q(x)]$ ← True

Which of the following is true?

(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) Neither S_1 nor S_2

meaning of Implies

$S_2: \exists_x\,(P(x) \rightarrow q(x))$

for at least one x, if P(x) is true then q(x) shd be true

@x=3 P(3) is true but q(3) is True

Negation of a statement

Let p(x) and q(x) denote the following open statements.

p(x): $x^2 > 0$

q(x): x is odd

for the universe of all integers, determine the truth or falsity of each of the statement.

S_1: $\forall x\ [p(x) \rightarrow q(x)]$

S_2: $\exists x\ [p(x) \rightarrow q(x)]$

Which of the following is true?

(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) Neither S_1 nor S_2

Negation of a statement

Consider

- Actor (x) = x is an actor
- Smart (x) = x is smart

and the well-formed formula:

$\exists x$ (Actor(x) $\wedge$ Smart (x))

Choose the correct representation of above in English sentence.

(A) Some Actor is smart

(B) Some Actor is not smart

(C) All actors are smart

(D) All smart are actors

Negation of a statement

"Every player is liked by some coach"

(A) $\forall(x)$ [player (x) $\rightarrow$ $\exists y$ [coach (y) $\wedge$likes (y, x)]]

(B) $\forall(x)$ [player (x) $\rightarrow$ $\exists y$ [coach (y) $\rightarrow$ likes (y, x)]]

(C) $\exists$ (x) [player (x) $\rightarrow$ $\forall y$ [coach (y) $\rightarrow$ likes (y, x)]]

(D) $\exists$ (x) [player (x) $\rightarrow$ $\forall y$ [coach (y) $\wedge$ likes (y, x)]]

(Every player is liked by every coach)

$$\left(\forall_x \text{Player}(x) \rightarrow \forall y\left(\text{Coach}(y) \rightarrow \text{Likes}(y,x)\right)\right)$$

Negation of a statement

Consider the following statement

"There is exactly one apple".

Let G(x) : x is an apple.

Now consider the predicate logic statements:

I. $\exists x \text{ apple}(x) \wedge \forall_y (\text{apple}(y)) \Rightarrow x = y)$

II. $\exists x \text{ apple}(x)$

The correct representation in predicate logic is ?

(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I and II

1) $\underbrace{\exists_x \text{ apple}(x)}_{\text{(at least one } x \text{ is apple)}} \wedge \underbrace{\forall_y((\text{apple}(y) \rightarrow (x=y))}_{\text{This will be true when apple(y) implies that } y=x}$

AND

$\underbrace{\exists_x \text{ apple}(x)}_{x \text{ is an apple}}$ $\wedge$ $\forall_y(\text{apple}(y) \rightarrow x=y)$

At least one x should be apple

$\forall_y(y \text{ is an apple} \xrightarrow{\text{should implies}} x=y)$

- $\forall x \forall y P(x,y) \Rightarrow$ This will be true if $P(x,y)$ is true for all pair of (x,y)

$\rightarrow$ It become false if $P(x,y)$ is for for any single pair (x,y)

$\equiv$

$\forall y \forall x P(x,y)$

equivalent.

- $\forall x \exists y P(x,y) \Rightarrow$ For all x there exist atleast one y where $P(x,y)$ is true.

- $\exists y \forall x P(x,y) \Rightarrow$ For at least one y the $P(x,y)$ is true for all x.

$\Rightarrow P(x,y) \Rightarrow (x+y=0)$ $(x=-y) \Rightarrow \forall x \exists y P(x,y)$ True

So I fix x and check

$x=3,\ y=-3$

$x=5,\ y=-5$

$\forall x \forall y P(x,y)$ False

$\forall x \forall y (x+y=0)$

$(3,3) \neq 0$

- $\forall x \exists y P(x,y) \Rightarrow$ For all x there exist atleast one y where $P(x,y)$ is true.

$\not\Rightarrow$

- $\exists y \forall x P(x,y) \Rightarrow$ For at least one y the $P(x,y)$ is true for all x.

$\Rightarrow P(x,y) \Rightarrow (x+y=0)$ $(x=-y)$

Take any one value of y → $y=100$ ⇒ Kya wo sabhi° x ko satisfy kar paegi

No

- $\forall x \exists y P(x,y) \Rightarrow$ For all x there exist atleast one y where $P(x,y)$ is true.

$$\not\Rightarrow$$

- $\exists y \forall x P(x,y) \Rightarrow$ For at least one y the $P(x,y)$ is true for all x.

if $\forall x \exists y P(x,y)$ is True then $\exists y \forall x P(x,y)$ is also True $\Rightarrow$ (False)

if $\exists y \forall x P(x,y)$ is true then $\forall x \exists y P(x,y)$ is also True $\Rightarrow$ True

$$\Rightarrow \exists y \forall x P(x,y) \longrightarrow \forall x \exists y P(x,y)$$

But not vice versa

Similarly

$$\Rightarrow \quad \exists_x \forall_y P(x,y) \longrightarrow \forall_y \exists_x P(x,y)$$

But not vice versa

- $\exists_x \exists_y P(x,y)$ is true if $P(x,y)$ is true for at least one pair of (x,y).

$$\equiv$$

$$\exists_y \exists_x P(x,y)$$

$$\forall_x \forall_y P(x,y) \equiv \forall_y \forall_x P(x,y)$$

$$\exists_x \exists_y P(x,y) \equiv \exists_y \exists_x P(x,y)$$

$$\exists_y \forall_x P(x,y) \rightarrow \forall_x \exists_y P(x,y)$$

$$\exists_x \forall_y P(x,y) \rightarrow \forall_y \exists_x P(x,y)$$

1) $\equiv$: $P \equiv Q$ if P is T then Q is True, vv
if P is F then Q is false, vv

2) $\longrightarrow$: $P \rightarrow Q$
if P then Q
if $P = T$ then $Q = T$

[MCQ]

Which of the following is/are true?

A. $\forall x\, \exists y\, P(x, y) \Rightarrow \exists y\, \forall x\, P(x, y)$

B. $\forall x\, \forall y\, P(x, y) \equiv \exists y\, \forall x\, P(x, y)$

C. $\exists x\, \forall y\, P(x, y) \Rightarrow \forall y\, \exists x\, P(x, y)$

D. $\exists x\, \exists y\, P(x, y) \Rightarrow \forall y\, \exists x\, P(x, y)$

True $\boxed{\forall y\, \exists x\, P(x,y) \longrightarrow \exists y\, \exists x\, P(x,y)}$

True $\boxed{\forall x\, \exists y\, P(x,y) \longrightarrow \exists y\, \exists x\, P(x,y)}$

[NAT]

Consider the first order logic sentence $\boxed{F : \forall x(\exists y\ R(x, y))}$. Assuming non-empty logical domains, which of the sentences below are implied by F?

I. $\exists y(\exists x\ R(x, y))$

II. $\exists y(\forall x\ R(x, y))$

III. $\forall y(\exists x\ R(x, y))$

IV. $\sim \exists x(\forall y \sim R(x, y))$

$\forall x\ \exists y\ (R(x,y))$

$\sim\sim\ \forall x\ \exists y\ (R(x,y))$

$\sim \exists x\ \forall y\ (\sim R(x,y))$

Relationship diagram

$$
\begin{tikzcd}
 & \forall x \forall y \arrow[rr, leftrightarrow] \arrow[dl] \arrow[ddl, dashed] \arrow[ddr, dashed] & & \forall y \forall x \arrow[dr] \arrow[ddl, dashed] \arrow[ddr, dashed] & \\
\exists x \forall y \arrow[d] \arrow[ddr, dashed] & & & & \exists y \forall x \arrow[d] \arrow[ddl, dashed] \\
\forall y \exists x \arrow[dr] & & & & \forall x \exists y \arrow[dl] \\
 & \exists y \exists x \arrow[rr, leftrightarrow] & & \exists x \exists y &
\end{tikzcd}
$$

Predicate Logic

. **Examples of Predicate Logic Statements**

- Universal Example:

  `∀x (Human(x) → Mortal(x))`

  For all x, If x is a human then x is mortal

- Existential Example:

  `∃x (Human(x) ∧ Rich(x))`

  For some x, x is human and Rich.

- More Complex Example:

  `∀x ∃y (Parent(y, x) ∧ Loves(y, x))`

  y is Parent of x
  y love x

  → For all x, there exist some y who are Parents of x and love x

• All Parent love their Children

$\forall y\ (Parent(y) \rightarrow loves(y,x))$

sentences with multiple quantifiers

Quantifications of Two Variables		
Statement	***When True?***	***When False?***
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y.
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y.

. $\forall x \exists y (x + y = 0)$ is read as:

a. For all x and y, $x + y = 0$

b. For all y, there exists an x such that $x + y = 0$

c. For all x, there exists a y such that $x + y = 0$

d. For all x and y, $xy = 0$

Express the statement "for every x and for every y, $x + y > 10$"

a. $\forall x \forall y P(y, y)$

b. $\forall x \forall y P(x, y)$

c. $\forall y P(x > y)$

d. $\forall x \forall y P(x)$

sentences with multiple quantifiers

done

$\exists x\ \forall y\ (P(x, y)) \Rightarrow \forall y\ \exists x\ (P(x, y))$

- **Interpretation of $\exists x\ \forall y\ (P(x, y))$**: There exists an **x** such that for all **y**, the property **P(x, y)** holds. This means there is a single **x** that works for all **y**.

- **Interpretation of $\forall y\ \exists x\ (P(x, y))$**: For every **y**, there exists an **x** such that the property **P(x, y)** holds. This means that for each **y**, we can find some **x**, but the **x** could be different for different values of **y**.

sentences with multiple quantifiers

done

- $\forall y\ \exists x\ (P(x, y)) \Leftarrow \exists x\ \forall y\ (P(x, y))$ (reverse implication)
- However, the reverse is not true.
- $\forall y\ \exists x\ (P(x, y))$ says that for each y, we can find an x such that P(x, y) holds, but it doesn't guarantee that there is one single x that works for all y.
- In other words, different values of y might require different values of x. Therefore, $\forall y\ \exists x\ (P(x, y))$ does not imply $\exists x\ \forall y\ (P(x, y))$.

sentences with multiple quantifiers

done

- Let P(x, y) be "x + y = 0".
- ∃x ∀y P(x, y): There exists a single x such that for all y, x + y = 0. This is false because no single x satisfies the equation for all y.
- ∀y ∃x P(x, y): For every y, there exists some x such that x + y = 0. This is true because for any y, we can choose x = -y, but the value of x depends on y.
- Thus, ∀y ∃x (P(x, y)) holds, but ∃x ∀y (P(x, y)) does not, showing that the reverse implication doesn't hold.

sentences with multiple quantifiers

done

- $\forall x \forall y P(x, y)$:

sentences with multiple quantifiers

- $\exists x \exists y P(x, y)$:

done

sentences with multiple quantifiers

1. ∀x ∃y (Loves(x, y))

For all x there exist some y who love x

2. ∃x ∀y (Loves(x, y))

There exist at least one x that is loved by all y.

3. ∀x ∃y (Teaches(y, x))

for all x there exist at least one y that teaches x

sentences with multiple quantifiers

4. ∃x ∀y (Friend(y, x))

   - "There exists a person x who is a friend to every person y."

5. ∀x ∀y (Person(x) ∧ Person(y) → ∃z (Friend(x, z) ∧ Friend(y, z)))

   for all x and y, if x and y are person then there is some z who is friend of x and y both.

6. ∀x ∃y ∀z (Knows(x, z) → Knows(y, z))

   for all x and z, if z knows x then for some y z knows y.

Rule of inference w.r.t Predicate Logic

- In predicate logic, rules of inference are used to derive conclusions from premises that contain quantifiers (like ∀ for "for all" and ∃ for "there exists"). These rules are extensions of propositional logic rules to handle predicates and quantifiers.
- Here are some key **rules of inference** in **predicate logic**:

Rule of inference w.r.t Predicate Logic

- 1. Universal Instantiation (UI):
- This rule allows us to derive a specific instance from a universally quantified statement.
- General Form: $\forall x\, P(x)$ implies $P(c)$
- If a property P(x) holds for all x, then it holds for any specific c.

Rule of inference w.r.t Predicate Logic

- Universal Generalization (UG):
- This rule allows us to generalize from a specific case to a universal statement.
- General Form: $P(c)$ implies $\forall x\, P(x)$
- If P(c) is true for an arbitrary c, then $P(x)$ is true for all x.

Rule of inference w.r.t Predicate Logic

- Existential Instantiation (EI):
- This rule allows us to introduce a specific instance from an existentially quantified statement.
- General Form: $\exists x\, P(x)$ implies $P(c)$
- If P(x) holds for some x, then it holds for a specific c, where c is a new symbol that doesn't appear anywhere else.

Rule of inference w.r.t Predicate Logic

- Existential Generalization (EG):
- This rule allows us to conclude an existential statement from a specific instance.
- General Form: $P(c)$ implies $\exists x\, P(x)$
- If P(c) is true for a specific c, then there exists some x such that P(x) is true.

Rule of inference w.r.t Predicate Logic

- Equivalence and Implications

THANK - YOU